EXPERIMENT-9

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Design input forms that validate data and display error messages using HTML/CSS,JavaScript (with Validator.js)

Aim:

To design and implement a registration form that validates user input such as name, email, phone number, and password using HTML, CSS, and JavaScript.

Procedure:

1. Created three files: index.html, style.css, and script.js.
2. Designed a registration form using HTML with fields like name, email, phone, password, and confirm password.
3. Styled the form using CSS to make it neat and readable.
4. Wrote JavaScript code to check the input values.
5. Displayed error messages for invalid inputs.

6. Prevented form submission if any field is incorrect.

7. Run the program by opening index.html in the browser.

Program:

1.Index.html:

index.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Professional Login</title>
    <!-- Google Fonts for Modern Typography -->
    <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600;700&display=swap" rel="stylesheet">
    <!-- FontAwesome for Icons -->
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <!-- Background Animated Elements -->
    <div class="background">
        <div class="shape"></div>
        <div class="shape"></div>
    </div>

    <!-- Login Container -->
    <div class="container">
        <div class="login-content">
            <div class="login-header">
                <h2>Welcome Back</h2>
                <p>Please login to your account.</p>
            </div>

            <!-- Login Form -->
            <form id="loginForm" class="form-card">

                <!-- Email Input -->
                <div class="input-group">
                    <div class="input-icon">
                        <i class="fas fa-envelope"></i>
                    </div>
                    <input type="text" id="email" placeholder="Email Address" autocomplete="off">
                </div>
                <span id="emailError" class="error-text"></span>

                <!-- Password Input -->
                <div class="input-group">
                    <div class="input-icon">
                        <i class="fas fa-lock"></i>
                    </div>
                    <input type="password" id="password" placeholder="Password">
```

```html
<input type="password" id="password" placeholder="Password">
<div class="password-toggle" id="togglePassword">
    <i class="far fa-eye"></i>
</div>
</div>
<span id="passwordError" class="error-text"></span>

<!-- Options -->
<div class="form-options">
    <label class="remember-me">
        <input type="checkbox" id="remember">
        <span>Remember me</span>
    </label>
    <a href="#" class="forgot-password">Forgot Password?</a>
</div>

<!-- Submit Button -->
<button type="submit" class="login-btn" id="loginBtn">
    <span>Login</span>
    <i class="fas fa-arrow-right"></i>
</button>

<!-- Social Login -->
<div class="social-login">
    <p>Or continue with</p>
    <div class="social-icons">
        <button type="button" class="social-btn"><i class="fab fa-google"></i></button>
        <button type="button" class="social-btn"><i class="fab fa-apple"></i></button>
        <button type="button" class="social-btn"><i class="fab fa-facebook-f"></i></button>
    </div>
</div>

<!-- Register Link -->
<div class="register-link">
    <p>Don't have an account? <a href="#">Sign up now</a></p>
</div>
```

```html
                <button type="button" class="social-btn"><i class="fab fa-google"></i></button>
                <button type="button" class="social-btn"><i class="fab fa-apple"></i></button>
                <button type="button" class="social-btn"><i class="fab fa-facebook-f"></i></button>
            </div>
        </div>

        <!-- Register Link -->
        <div class="register-link">
            <p>Don't have an account? <a href="#">Sign up now</a></p>
        </div>

    </form>
  </div>
</div>

<script src="script.js"></script>
</body>
</html>
```

2.Style.css:

style.css > :root

```css
:root {
    --primary-color: #6366f1;
    --primary-hover: #4f46e5;
    --bg-gradient-1: #0f172a;
    --bg-gradient-2: #1e1b4b;
    --text-main: #f8fafc;
    --text-muted: #94a3b8;
    --input-bg: rgba(255, 255, 255, 0.05);
    --input-border: rgba(255, 255, 255, 0.1);
    --input-focus: rgba(99, 102, 241, 0.5);
    --card-bg: rgba(15, 23, 42, 0.6);
    --card-border: rgba(255, 255, 255, 0.08);
}

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: 'Poppins', sans-serif;
}

body {
    background: linear-gradient(135deg, var(--bg-gradient-1), var(--bg-gradient-2));
    min-height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    color: var(--text-main);
    overflow: hidden;
    position: relative;
}

/* Background animated shapes */
.background {
    position: absolute;
    width: 100vw;
    height: 100vh;
    top: 0;
    left: 0;
    z-index: 0;
    overflow: hidden;
}

.shape {
    position: absolute;
```

```
.shape {
    position: absolute;
    filter: blur(90px);
    z-index: 0;
    animation: float 10s infinite alternate;
}

.shape:nth-child(1) {
    width: 450px;
    height: 450px;
    background: #6366f1;
    top: -100px;
    left: -100px;
    border-radius: 50%;
    opacity: 0.4;
}

.shape:nth-child(2) {
    width: 500px;
    height: 500px;
    background: #ec4899;
    bottom: -150px;
    right: -100px;
    border-radius: 50%;
    opacity: 0.3;
    animation-delay: -5s;
}

@keyframes float {
    0% { transform: translateY(0) scale(1); }
    100% { transform: translateY(30px) scale(1.1); }
}

/* Login Container */
.container {
    position: relative;
    z-index: 1;
    width: 100%;
    max-width: 420px;
    padding: 20px;
}

.login-content {
    background: var(--card-bg);
```

```
    -webkit-backdrop-filter: blur(20px);
    border: 1px solid var(--card-border);
    border-radius: 24px;
    padding: 40px;
    box-shadow: 0 25px 50px -12px rgba(0, 0, 0, 0.4);
    animation: slideUp 0.6s cubic-bezier(0.16, 1, 0.3, 1);
}

@keyframes slideUp {
    from { opacity: 0; transform: translateY(30px); }
    to { opacity: 1; transform: translateY(0); }
}

.login-header {
    text-align: center;
    margin-bottom: 35px;
}

.login-header h2 {
    font-size: 32px;
    font-weight: 600;
    margin-bottom: 8px;
    background: linear-gradient(to right, #fff, #cbd5e1);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
}

.login-header p {
    color: var(--text-muted);
    font-size: 14px;
}

/* Form inputs */
.form-card {
    display: flex;
    flex-direction: column;
}

.input-group {
    position: relative;
    margin-bottom: 5px;
    display: flex;
    align-items: center;
```

```
.input-icon {
    position: absolute;
    left: 18px;
    color: var(--text-muted);
    font-size: 18px;
    transition: color 0.3s;
    pointer-events: none;
}

input {
    width: 100%;
    padding: 16px 45px 16px 50px;
    border: 1px solid var(--input-border);
    background: var(--input-bg);
    border-radius: 12px;
    color: var(--text-main);
    font-size: 15px;
    transition: all 0.3s;
    outline: none;
}

input::placeholder {
    color: rgba(203, 213, 225, 0.5);
}

input:focus {
    border-color: var(--input-focus);
    background: rgba(255, 255, 255, 0.08); /* Slightly brighter on focus */
    box-shadow: 0 0 0 4px rgba(99, 102, 241, 0.15);
}

/* Color icon on input focus/content */
.input-group:focus-within > .input-icon,
input:not(:placeholder-shown) ~ .input-icon {
    color: var(--primary-color);
}

.password-toggle {
    position: absolute;
    right: 18px;
    color: var(--text-muted);
    cursor: pointer;
    transition: color 0.3s;
```

```
    align-items: center;
}

.password-toggle:hover {
    color: var(--text-main);
}

.error-text {
    color: #ef4444;
    font-size: 12px;
    min-height: 20px;
    margin-bottom: 10px;
    margin
    displa
}

/* Shake A
@keyframes
    0%, 10
    25% { transform: translateX(-5px); }
    50% { transform: translateX(5px); }
    75% { transform: translateX(-5px); }
}

.shake {
    animation: shake 0.4s;
}

/* Options */
.form-options {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 25px;
    font-size: 13px;
}

.remember-me {
    display: flex;
    align-items: center;
    cursor: pointer;
    color: var(--text-muted);
}
```

A two-dimensional transformation is applied to an element through the 'transform' property. This property co
those allowed by SVG.

Widely available across major browsers (Baseline since 2015)

Syntax: none | <transform-list>

MDN Reference

```
    cursor: pointer;
    color: var(--text-muted);
}

.remember-me input {
    width: auto;
    margin-right: 8px;
    accent-color: var(--primary-color);
    cursor: pointer;
}

.forgot-password {
    color: var(--primary-color);
    text-decoration: none;
    font-weight: 500;
    transition: color 0.3s;
}

.forgot-password:hover {
    color: var(--text-main);
    text-decoration: underline;
}

/* Button */
.login-btn {
    display: flex;
    justify-content: center;
    align-items: center;
    gap: 10px;
    width: 100%;
    padding: 16px;
    border: none;
    background: var(--primary-color);
    color: white;
    font-size: 16px;
    font-weight: 600;
    border-radius: 12px;
    cursor: pointer;
    transition: all 0.3s;
    box-shadow: 0 4px 15px rgba(99, 102, 241, 0.4);
}

.login-btn:hover {
    background: var(--primary-hover);
```

```
.login-btn:hover {
    background: var(--primary-hover);
    transform: translateY(-2px);
    box-shadow: 0 6px 20px rgba(99, 102, 241, 0.6);
}

.login-btn:active {
    transform: translateY(0);
}

/* Social Login */
.social-login {
    margin-top: 30px;
    text-align: center;
}

.social-login p {
    color: var(--text-muted);
    font-size: 13px;
    margin-bottom: 20px;
    position: relative;
}

.social-login p::before,
.social-login p::after {
    content: '';
    position: absolute;
    top: 50%;
    width: 25%;
    height: 1px;
    background: var(--input-border);
}

.social-login p::before { left: 0; }
.social-login p::after { right: 0; }

.social-icons {
    display: flex;
    justify-content: center;
    gap: 15px;
}

.social-btn {
    display: flex;
```

```
.social-btn {
    display: flex;
    justify-content: center;
    align-items: center;
    width: 45px;
    height: 45px;
    border-radius: 50%;
    border: 1px solid var(--input-border);
    background: var(--input-bg);
    color: var(--text-main);
    font-size: 18px;
    cursor: pointer;
    transition: all 0.3s;
}

.social-btn:hover {
    background: rgba(255, 255, 255, 0.15);
    transform: translateY(-3px);
    border-color: rgba(255, 255, 255, 0.3);
    color: var(--primary-color);
}

/* Register link */
.register-link {
    text-align: center;
    margin-top: 25px;
    font-size: 14px;
    color: var(--text-muted);
}

.register-link a {
    color: var(--primary-color);
    text-decoration: none;
    font-weight: 500;
    transition: color 0.3s;
}

.register-link a:hover {
    color: var(--text-main);
    text-decoration: underline;
}
```

3.Srcipt.js:

script.js > document.addEventListener('DOMContentLoaded') callback > togglePassword.addEventListener('click') callback

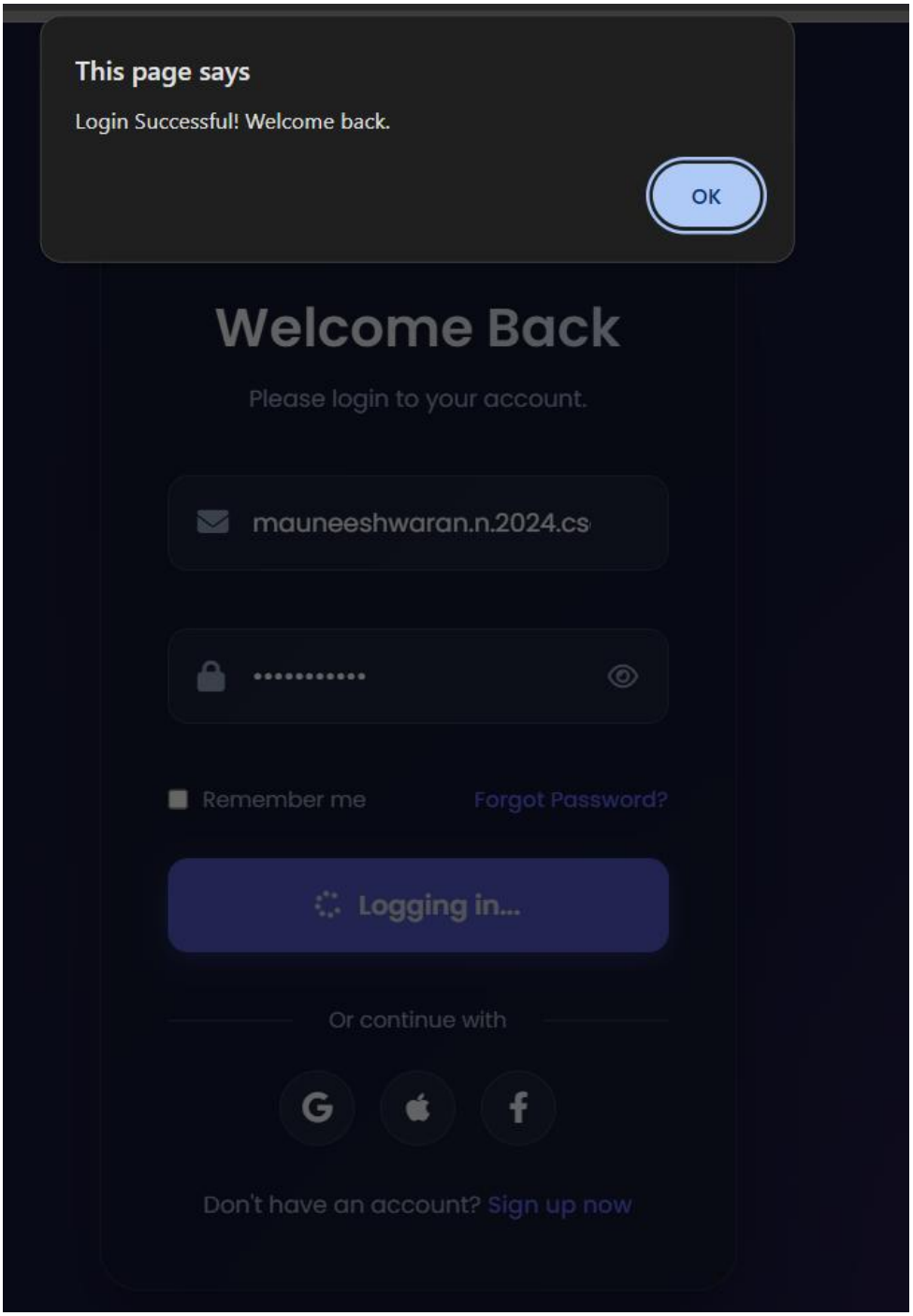
```javascript
document.addEventListener('DOMContentLoaded', () => {
    const loginForm = document.getElementById('loginForm');
    const emailInput = document.getElementById('email');
    const passwordInput = document.getElementById('password');
    const emailError = document.getElementById('emailError');
    const passwordError = document.getElementById('passwordError');
    const togglePassword = document.getElementById('togglePassword');
    const loginBtn = document.getElementById('loginBtn');

    // Toggle Password Visibility
    togglePassword.addEventListener('click', function() {
        const type = passwordInput.getAttribute('type') === 'password' ? 'text' : 'password';
        passwordInput.setAttribute('type', type);

        // Toggle icon
        const icon = this.querySelector('i');
        if (type === 'text') {
            icon.classList.remove('fa-eye');
            icon.classList.add('fa-eye-slash');
        } else {
            icon.classList.remove('fa-eye-slash');
            icon.classList.add('fa-eye');
        }
    });

    // Form Submit Event
    loginForm.addEventListener('submit', (e) => {
        e.preventDefault();

        let isValid = true;
        const email = emailInput.value.trim();
        const password = passwordInput.value;

        // Reset errors and styles
        emailError.textContent = '';
        passwordError.textContent = '';
        emailInput.style.borderColor = '';
        passwordInput.style.borderColor = '';

        // Validate Email
        if (!email) {
            showError(emailInput, emailError, 'Email is required');
            isValid = false;
        } else if (!isValidEmail(email)) {
```

```javascript
        // Validate Email
        if (!email) {
            showError(emailInput, emailError, 'Email is required');
            isValid = false;
        } else if (!isValidEmail(email)) {
            showError(emailInput, emailError, 'Please enter a valid email');
            isValid = false;
        }

        // Validate Password
        if (!password) {
            showError(passwordInput, passwordError, 'Password is required');
            isValid = false;
        } else if (password.length < 6) {
            showError(passwordInput, passwordError, 'Password must be at least 6 characters');
            isValid = false;
        }

        // Simulate Login Process if Valid
        if (isValid) {
            // Apply loading state to button
            const originalBtnContent = loginBtn.innerHTML;
            loginBtn.innerHTML = '<i class="fas fa-spinner fa-spin"></i> <span>Logging in...</span>';
            loginBtn.disabled = true;
            loginBtn.style.opacity = '0.8';
            loginBtn.style.cursor = 'not-allowed';

            // Simulate network request
            setTimeout(() => {
                // Remove loading state
                loginBtn.innerHTML = originalBtnContent;
                loginBtn.disabled = false;
                loginBtn.style.opacity = '1';
                loginBtn.style.cursor = 'pointer';

                alert('Login Successful! Welcome back.');
                loginForm.reset();
            }, 1500);
        }
    });
```

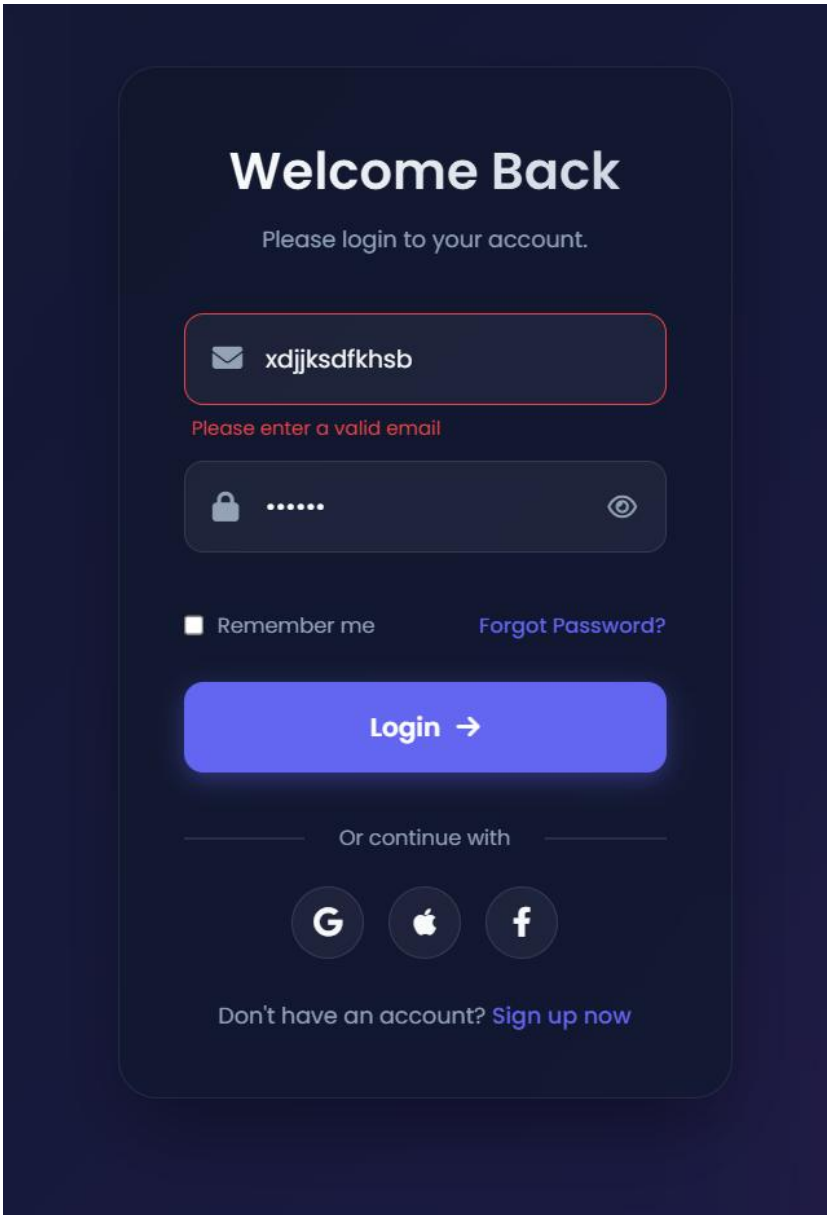
```javascript
document.addEventListener('DOMContentLoaded', () => {
    loginForm.addEventListener('submit', (e) => {
        setTimeout(() => {
            loginBtn.style.opacity = '1';
            loginBtn.style.cursor = 'pointer';

            alert('Login Successful! Welcome back.');
            loginForm.reset();
        }, 1500);
    }
});

// Helper functions
function showError(inputElement, errorElement, message) {
    errorElement.textContent = message;
    inputElement.style.borderColor = '#ef4444'; // Red border

    // Trigger generic shake animation
    const inputGroup = inputElement.parentElement;
    inputGroup.classList.remove('shake');
    // Force reflow
    void inputGroup.offsetWidth;
    inputGroup.classList.add('shake');
}

function isValidEmail(email) {
    const re = /^(([^<>()\[\]\\.,;:\s@"]+(\.[^<>()\[\]\\.,;:\s@"]+)*)|(".+"))@((\[[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}])|(([a-zA-Z\-0-9]+\.)+[a-zA-Z]{2,}))$/
```

Output:

1.For valid mail:



2.For invalid mail:



Result:

The login page successfully validates user inputs and displays appropriate error messages for incorrect entries.